MYLOCALASSISTANT

Tester VM Setup Guide

A complete, step-by-step guide for renting and configuring a cloud virtual machine to test the application — no prior technical experience required.

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1. Overview – What Is a VM and Why Do You Need One?

A **Virtual Machine (VM)** is a computer that runs inside another computer, hosted in a data centre somewhere in the world. When you rent a VM from a cloud provider like Amazon Web Services (AWS), you get access to a full Windows computer — with its own processor, memory, storage, and in our case a powerful graphics card (GPU) — that you can control over the internet.

You connect to it using **Remote Desktop Connection**, a built-in Windows application that lets you see and interact with the VM's screen from your own PC, just like sitting in front of it.

Why a VM and not just my own PC?

MyLocalAssistant runs AI models locally on the machine. To run even medium-sized AI models at a usable speed, you need a **dedicated graphics card (GPU) with at least 8–16 GB of video memory (VRAM)**. Most standard laptops and office desktops do not have this. A cloud VM gives you access to professional-grade NVIDIA GPUs for a small hourly fee — you only pay while you are actively testing.

What will you be doing?

- Create an AWS account and rent a Windows VM with a GPU
- Install the necessary software on the VM (GPU drivers, CUDA, .NET runtime)
- Copy the application from your local PC to the VM
- Run and test the application — local AI inference, cloud AI providers (OpenAI, Groq, etc.), RAG document search, agents, and tool calls
- Stop the VM when you are done to avoid unnecessary charges

> **! BEFORE YOU START**
>
> You will need a **credit or debit card** to sign up for AWS. AWS charges by the hour, but **only while the VM is actively running**. You will not be charged while the VM is stopped. Always stop the VM when you finish a testing session.

Key terms explained

Term	Plain-English meaning
VM (Virtual Machine)	A cloud computer you rent and control over the internet.
GPU (Graphics Processing Unit)	A specialised chip that runs AI model calculations very fast. Essential for local AI inference.
VRAM (Video RAM)	Memory on the GPU. The bigger the AI model, the more VRAM it needs.
CUDA	NVIDIA's software platform that allows programs to use the GPU. Required for AI acceleration.
RDP (Remote Desktop Protocol)	The technology that lets you control the VM's screen from your PC.
EC2 (Elastic Compute Cloud)	The name AWS uses for their virtual machines.
Instance	AWS term for a single running VM.

2. VM Options and Pricing

The table below lists the recommended cloud VM options. All prices are approximate on-demand (pay-per-hour) rates as of May 2026 and can vary slightly by region.

Provider	Instance Name	GPU Model	GPU VRAM	System RAM	CPU Cores	Approx. Cost/hr
AWS	g4dn.xlarge Recommended	NVIDIA Tesla T4	16 GB	16 GB	4 vCPU	**~$0.53**
AWS	g5.xlarge	NVIDIA A10G	24 GB	16 GB	4 vCPU	**~$1.01**
Azure	NC4as T4 v3	NVIDIA Tesla T4	16 GB	28 GB	4 vCPU	**~$0.53**
Vast.ai	RTX 4090 rental	NVIDIA RTX 4090	24 GB	32 GB	8–16	**~$0.35–0.60**

Recommendation: AWS g4dn.xlarge

The **g4dn.xlarge** offers the best balance of cost, ease of setup, and compatibility. It runs Windows Server 2022 natively (required for our application), has an NVIDIA T4 GPU with 16 GB VRAM which fits 13B parameter AI models comfortably, and costs about **$0.53 per hour** — roughly $2.10 for a 4-hour testing session.

! VAST.AI NOTE

Vast.ai offers cheaper GPU rentals but runs Linux, not Windows. Our application requires Windows. Do not use Vast.ai unless you are an advanced user testing only the server component.

What do these numbers actually mean?

- **GPU VRAM:** The most important number. A 7B parameter AI model at standard quality (Q4) needs ~5 GB. A 13B model needs ~8–10 GB. A 70B model needs ~40 GB. The T4 with 16 GB handles 13B models easily.
- **System RAM (16–28 GB):** Regular computer memory used by Windows, the app, and supporting processes. 16 GB is the comfortable minimum.
- **vCPU:** Virtual CPU cores. AI inference runs mainly on the GPU, so 4 cores is more than sufficient.

- **On-demand pricing:** You pay only for the hours the VM is in the “Running” state. Stopping it stops the charge.

3. Creating an AWS Account

Amazon Web Services (AWS) is the cloud platform where you will rent your VM. If you already have an AWS account, skip to Section 4.

1

Open the AWS sign-up page

Open any web browser on your PC and navigate to: `https://aws.amazon.com`

Click the orange **"Create an AWS Account"** button in the top-right corner of the page.

2

Enter your email address and choose a password

Enter your email address in the **"Root user email address"** field. Choose a name for the account (e.g. `mycompany-testing`). Set a strong password.

Click **"Verify email address"**. AWS will send a 6-digit verification code to your email inbox. Check your email and enter the code on the next screen, then click **"Verify"**.

3

Enter your contact information

Select **"Personal — for your own projects"** as the account type (unless your company is paying). Fill in your full name, phone number, country, and address. Click **"Continue"**.

4

Add a payment method

Enter your credit or debit card number, expiry date, and the security code. AWS will make a temporary small charge (typically $1) to verify the card is valid. This is automatically refunded within a few business days.

Click **"Verify and Continue"**.

> **! NOTE ON CHARGES**
>
> You will only be billed for actual usage. There are no monthly subscription fees. Charges appear only when a VM is running.

5

Complete phone verification

AWS will either call or send a text message to your phone number with a PIN code. Enter the PIN on the screen and click **"Continue"**.

6

Choose a support plan

Select **“Basic support — Free”**. This is the free tier and is perfectly adequate for testing purposes. Click **“Complete sign up”**.

7

Sign in to the AWS Management Console

Go to: `https://console.aws.amazon.com`

Select **“Root user”**, enter your email address and password, and click **“Sign in”**.

> **OK – YOU NOW HAVE AN AWS ACCOUNT**
>
> You are inside the AWS Management Console — a web dashboard from which you manage all your cloud resources. It may look complex; the next section walks you through exactly what to click.

> **TIP – SET A BILLING ALERT NOW (RECOMMENDED)**
>
> To avoid unexpected charges, set up a cost alert: in the AWS Console, click your account name (top-right) → **Billing and Cost Management** → **Budgets** → **Create budget**. Set a monthly budget of e.g. $30 and AWS will email you when your spending approaches that amount.

4. Renting a VM on AWS (Step by Step)

In AWS, a virtual machine is called an **EC2 instance**. Follow the steps below to launch one.

1

Navigate to EC2

In the AWS Console, locate the search bar at the very top of the page (it says "Search"). Type `EC2` and click **"EC2"** in the results dropdown that appears below the search bar.

2

Select a region close to you

In the top-right corner of the AWS Console you will see a region name (e.g. `Ohio` or `Ireland`). Click it to open the region list and choose one geographically close to you for best performance. **US East (N. Virginia)** or **EU (Ireland)** are reliable choices.

> **TIP – REGION AVAILABILITY**
>
> GPU instances (g4dn) are not available in every region. If you encounter availability issues in step 6, try switching to **US East (N. Virginia)**.

3

Click "Launch instance"

On the EC2 dashboard page, click the orange **"Launch instance"** button. This opens the instance configuration wizard.

4

Name your instance

In the **"Name and tags"** section at the top, type a name for your VM. For example: `MyLocalAssistant-Test`. This name is just a label to help you identify it later.

5

Choose the operating system

Under **"Application and OS Images (Amazon Machine Image)"**, click **"Browse more AMIs"**.

In the search box that appears, type `Windows Server 2022` and press Enter.

Find **"Microsoft Windows Server 2022 Base"** in the list and click **"Select"** next to it.

> **TIP – WHY WINDOWS?**
>
> Our application uses Windows Forms technology and only runs on Windows. Windows Server 2022 is the most stable and widely available Windows option on AWS.

6 Choose the instance type (hardware)

Under **"Instance type"**, click inside the search field and type `g4dn.xlarge`. Select it from the list.

Verify you can see: **4 vCPU**, **16.0 GiB memory**. This is the GPU instance with the NVIDIA T4.

> **! COST REMINDER**
>
> This instance costs approximately $0.53 per hour. A 4-hour session costs around $2.10. Remember to stop the instance when you are done testing (see Section 10).

7 Create a key pair (needed to retrieve the password)

Under **"Key pair (login)"**, click **"Create new key pair"**. A dialog box appears:

- **Key pair name:** type `myassistant-test-key`
- **Key pair type:** RSA
- **Private key file format:** .pem

Click **"Create key pair"**. Your browser will automatically download a file named `myassistant-test-key.pem`.

> **! SAVE THIS FILE – YOU CANNOT GET IT AGAIN!**
>
> Move the `myassistant-test-key.pem` file to a safe location on your PC (e.g. your Documents folder). You need it every time you want to retrieve the VM's password. If you lose it, you cannot log into the VM.

8 Configure network security (allow Remote Desktop access)

Under **"Network settings"**, click **"Edit"** (top-right of that section).

You should see an existing rule for **RDP, TCP port 3389**. Under **"Source type"** for this rule, select **"My IP"**. AWS will automatically fill in your current IP address. This ensures only you can connect.

> **TIP – IF YOUR IP CHANGES**
>
> If you cannot connect via RDP in the future, go to EC2 → Security Groups → find your group → edit the inbound rule for port 3389 and update the IP. Your current IP can be found at `https://whatismyipaddress.com`

9 Increase storage size

Under **"Configure storage"**, change the disk size from the default (usually 30 GiB) to **60 GiB**. This provides enough space for the application, the .NET runtime, GPU drivers, CUDA toolkit, and AI model files.

Leave the type as **gp3**.

10 Review and launch

Check the **"Summary"** panel on the right side of the screen. Confirm:

- Number of instances: **1**
- AMI: **Windows Server 2022**
- Instance type: **g4dn.xlarge**

Click the orange **"Launch instance"** button. AWS will display a success message. Click **"View all instances"** to return to your instance list.

11 Wait for the instance to start

In the Instances list, find your new instance. The **"Instance state"** column will initially show `Pending`. Wait approximately 3–5 minutes and refresh the page (press **F5**) until the state changes to `Running`.

> **OK – YOUR VM IS NOW RENTED AND RUNNING**
>
> The hourly charge begins now. Proceed to the next section to connect to it.

5. Connecting to Your VM via Remote Desktop

Remote Desktop Connection is a built-in Windows application that lets you see and control another computer over the internet. The steps below explain how to retrieve the VM's password and connect.

Step A — Retrieve the VM password

1

Open your instance details

In the EC2 Instances list, click on the name of your instance (`MyLocalAssistant-Test`) to open its detail panel.

2

Open the Connect dialog

Click the **"Connect"** button (orange, near the top of the page).

In the dialog that opens, click the **"RDP client"** tab at the top of the dialog.

3

Decrypt the password using your key file

Click **"Get password"**.

On the next screen, click **"Browse"** and select the `myassistant-test-key.pem` file you saved in Section 4, Step 7.

Click **"Decrypt password"**. A long password will appear. Click the copy icon next to the password to copy it to your clipboard.

> **! SAVE THIS PASSWORD**
>
> Paste the password into a text file or password manager right now. You will need it every time you connect. The password does not change unless you manually reset it inside the VM.

4

Download the RDP connection file

Still on the RDP client tab, click **"Download remote desktop file"**. Save the `.rdp` file to your Desktop. You can reuse this file every time you connect.

Step B — Connect

5 Open the RDP file

On your own PC, double-click the `.rdp` file you just downloaded.

A security warning will appear saying the remote computer's identity cannot be verified. This is expected for a new VM. Click **"Connect"** (or **"Yes"**) to proceed.

6 Enter the credentials

The **Windows Security** login dialog will appear:

- **Username:** `Administrator`
- **Password:** paste the password you copied in Step 3

Click **OK**.

7 Accept the certificate and wait for the desktop

A certificate warning may appear again. Click **"Yes"**.

After 5–15 seconds the VM's Windows desktop will appear inside your Remote Desktop window. You are now in control of the cloud computer.

> **OK – CONNECTED TO THE VM**
>
> Everything you do inside the Remote Desktop window happens on the cloud computer, not your local PC. Your local PC is just the viewer and keyboard/mouse.

> **TIP – FULL SCREEN**
>
> Press **Ctrl + Alt + Break** to toggle between full-screen and windowed Remote Desktop. The Break key may be labelled "Pause/Break" on your keyboard. Working in full screen is easier.

6. Installing Required Software on the VM

All steps in this section are performed **inside the VM** (i.e. inside the Remote Desktop window on your screen).

6.1 — .NET 8 Desktop Runtime

The application is built with Microsoft .NET 8. You need the "Desktop Runtime" version, which includes support for Windows Forms applications.

1. ### Open Microsoft Edge inside the VM

   Click the **Start button** (bottom-left of the VM's taskbar) and open **Microsoft Edge**. If it is not pinned, type "Edge" in the Start menu search box.

2. ### Download the .NET 8 Desktop Runtime

   In Edge, go to: `https://dotnet.microsoft.com/download/dotnet/8.0`

   Scroll down until you find the section labelled **".NET Desktop Runtime 8.x.x"**. It is important to choose this one — not the ".NET Runtime" and not the "SDK".

   Click **"Download x64"** (the 64-bit version).

3. ### Run the installer

   Open File Explorer inside the VM, navigate to the **Downloads** folder, and double-click the file whose name starts with `windowsdesktop-runtime-8`.

   Click **"Install"**. Wait until the progress bar completes. Click **"Close"**.

6.2 — NVIDIA GPU Driver

The GPU driver is a piece of software that allows Windows to communicate with and control the NVIDIA T4 graphics card. Without it the GPU cannot be used for AI inference.

4 Download the NVIDIA driver

In Edge on the VM, go to: `https://www.nvidia.com/Download/index.aspx`

Fill in the search form exactly as follows:

- **Product Type:** Data Center / Tesla
- **Product Series:** T-Series
- **Product:** Tesla T4
- **Operating System:** Windows Server 2022
- **Download Type:** Production Branch / Studio
- **Language:** English (US)

Click **"Search"**. A result will appear — click the **"Download"** button on it, then confirm by clicking **"Download"** again on the next page.

5 Install the driver

In the Downloads folder, double-click the NVIDIA installer file (its name starts with three numbers, e.g. `553.62-...`).

Choose **"Express (Recommended)"** when asked for the installation type. Click **"NEXT"** and wait for it to finish. If Windows asks you to restart, click **"RESTART NOW"**. After restarting, reconnect via Remote Desktop using the same .rdp file.

6.3 — CUDA Toolkit

CUDA is NVIDIA's platform for using the GPU to run general computations. The AI inference engine (LlamaSharp) requires CUDA to run AI models on the GPU. Without CUDA the app falls back to slow CPU inference.

6 Download CUDA Toolkit

In Edge, go to: `https://developer.nvidia.com/cuda-downloads`

Select the following options on the page:

- **Operating System:** Windows
- **Architecture:** x86_64
- **Version:** Server 2022
- **Installer Type:** exe (local)

Click **"Download"**. The file is large (approximately 2–4 GB) and may take 10–20 minutes to download.

7 Install CUDA Toolkit

Run the downloaded `.exe` file from the Downloads folder. Choose **"Express (Recommended)"** when prompted. The installation takes 5–15 minutes. Click **"Close"** when done. Restart the VM if asked.

6.4 — Verify the GPU is working

8

Test the GPU in PowerShell

Right-click the **Start button** and click **"Windows PowerShell"** (or **"Terminal"**). Type the following and press Enter:

```
nvidia-smi
```

If the installation was successful, you will see a table showing the Tesla T4 GPU, driver version, CUDA version, and memory usage. If you see an error, restart the VM and try again.

7. Preparing the Application on Your Local PC

This section is done on **your own PC**, not the VM. You will create a portable, self-contained copy of the application that can be run on the VM without needing the full development environment.

> **TIP – PREREQUISITE**
>
> Your local PC needs the .NET 8 SDK installed. To check, open PowerShell and type `dotnet --version` — it should show `8.x.x`.

1. ### Open PowerShell on your local PC

   Press **Windows key + X** on your keyboard and click **"Windows PowerShell"** or **"Terminal"** from the menu that pops up.

2. ### Navigate to the project folder

   Type the command below, replacing the path with the actual location of the project on your PC, and press Enter:

   ```
   cd "D:\VSCodeRepo\MyLocalAssistant"
   ```

3. ### Publish the Server Host application

   ```
   dotnet publish src\MyLocalAssistant.ServerHost -c Release -r win-x64 --self-contained true -o publish\ServerHost
   ```

   Wait for it to complete. It will print "Build succeeded" when done.

4. ### Publish the Client application

   ```
   dotnet publish src\MyLocalAssistant.Client -c Release -r win-x64 --self-contained true -o publish\Client
   ```

5. ### Publish the Admin application

   ```
   dotnet publish src\MyLocalAssistant.Admin -c Release -r win-x64 --self-contained true -o publish\Admin
   ```

6

Compress the publish folder into a ZIP file

Open **File Explorer** and navigate to the project folder. You will see a new folder called `publish` containing three subfolders: `ServerHost`, `Client`, and `Admin`.

Right-click the `publish` folder and:

- **Windows 11:** click **"Compress to ZIP file"**
- **Windows 10:** click **"Send to"** → **"Compressed (zipped) folder"**

Name the resulting file `MyLocalAssistant-publish.zip`. It will be approximately 150–350 MB in size.

> **OK – READY TO TRANSFER**
>
> The ZIP file contains everything needed to run the application on the VM. No additional .NET installation is required on the VM other than what you did in Section 6.

8. Transferring the Application to the VM

You need to copy the ZIP file from your local PC to the VM. The easiest way is **drive sharing via Remote Desktop**, which lets you access your local drives from inside the VM — no extra software needed.

Method A — Remote Desktop Drive Sharing (Recommended)

1

Open Remote Desktop Connection with options

On your local PC, press **Windows key + R**, type `mstsc` and press Enter. Click **"Show Options"** at the bottom-left of the window.

2

Enable local drive sharing

Click the **"Local Resources"** tab. Under **"Local devices and resources"**, click **"More..."**.

In the dialog that appears, tick the checkbox next to **"Drives"** (expand it and tick the drive where your ZIP file is located — usually C: or D:). Click **OK**.

3

Connect to the VM

Go back to the **"General"** tab. In the **"Computer"** field, enter the VM's Public IPv4 address (from the EC2 instance details page in the AWS Console). In **"User name"**, type `Administrator`.

Click **"Connect"** and enter the VM password when prompted.

4

Find your local drive inside the VM

Inside the Remote Desktop window, open **File Explorer** (press **Windows key + E**). In the left panel, click **"This PC"**. Scroll down to find the section called **"Redirected drives and folders"**. Your local PC's drives will appear there, labelled something like **"C on YOURPCNAME"**.

Click that drive, navigate to the `MyLocalAssistant-publish.zip` file, **copy it** (Ctrl+C), then **paste it** (Ctrl+V) onto the VM's Desktop or into `C:\MyLocalAssistant`.

5

Extract the ZIP file on the VM

Right-click the ZIP file in File Explorer on the VM and select **"Extract All..."**.

Set the destination to `C:\MyLocalAssistant` and click **"Extract"**. After extraction you will see three folders: `ServerHost`, `Client`, and `Admin`.

Method B — Cloud Storage (Alternative)

If drive sharing does not work, upload the ZIP to OneDrive, Google Drive, or Dropbox from your local PC. Then inside the VM, open Edge and log in to the same cloud storage service to download the file.

9. Running the Application

The application has three parts that must be started in the correct order:

1. **ServerHost** — starts the backend AI server (always start this first)
2. **Client** — the end-user chat interface
3. **Admin** — the administrator panel for managing users, agents, and settings

1

Start the Server Host

Inside the VM, open File Explorer and navigate to `C:\MyLocalAssistant\ServerHost\`.

Double-click `MyLocalAssistant.ServerHost.exe`. A console window will appear. Wait until you see a message like **"Application started. Press Ctrl+C to shut down."** This usually takes 10–30 seconds. **Do not close this window** — it must stay open while testing.

> **TIP – FIRST-RUN SETUP**
>
> On the very first run, the server creates its database and default admin account. The console window will display the default administrator username and a generated password. **Write these down** — you need them to log into the Client and Admin apps.

2

Start the Client (chat interface)

Open a new File Explorer window, navigate to `C:\MyLocalAssistant\Client\`, and double-click `MyLocalAssistant.Client.exe`.

The chat interface opens. Log in using the credentials displayed by the ServerHost in Step 1.

3

Start the Admin panel

Navigate to `C:\MyLocalAssistant\Admin\` and double-click `MyLocalAssistant.Admin.exe`.

Log in with the same admin credentials. From here you can manage users and agents, configure settings, view audit logs, and check usage statistics.

4

Download a local AI model (for local inference testing)

To test the application's ability to run AI locally on the GPU, you need to download a model. In the Admin panel, go to the **Models** tab and use the built-in downloader.

A good starting model for 16 GB VRAM (T4 GPU) is: **Llama-3.1-8B-Instruct-Q4_K_M.gguf** (approximately 5 GB). This downloads directly to the VM.

5

Configure cloud API keys (optional)

To test the cloud AI providers (OpenAI, Anthropic, Groq, Gemini, Mistral, Cerebras), go to Admin → **Settings** and enter API keys for the providers you want to test.

Free API keys with generous free tiers are available from:

- **Groq** (very fast, free tier): `https://console.groq.com`
- **Google Gemini** (free tier): `https://aistudio.google.com`
- **OpenAI** (paid): `https://platform.openai.com`

OK – READY TO TEST

With all three applications running you can now test: local AI inference (GPU), cloud AI providers, RAG document search, agents, tool calls, conversation history, and the admin management features.

10. Managing Your VM — Stopping and Starting

! CRITICAL – STOP THE VM WHEN YOU ARE DONE TESTING

A running g4dn.xlarge costs approximately $0.53 every hour. Leaving it on overnight accidentally costs ~$13. Always stop the VM when you finish a testing session.

How to STOP the VM (pauses it — all your data is preserved)

1. ### Close the Remote Desktop session first

   Click the **X** on the Remote Desktop window bar at the top of your screen. When asked, choose **"Disconnect"**. The VM keeps running until you stop it from the AWS Console.

2. ### Open the AWS Console and navigate to EC2

   On your local PC, go to `https://console.aws.amazon.com`, sign in, search for **EC2**, and click **"Instances"** in the left panel.

3. ### Stop the instance

   Tick the checkbox next to `MyLocalAssistant-Test`. Click the **"Instance state"** dropdown button near the top of the page. Select **"Stop instance"**. Confirm by clicking **"Stop"** in the dialog.

   The state changes to `Stopping`, then to `Stopped` within 1–2 minutes. GPU charges stop immediately.

TIP – STOPPED VS TERMINATED

Stopped: the VM is paused. Your files, installed software, downloaded models, and settings are all preserved. A small charge continues for disk storage (~$0.08/GB/month).

Terminated: the VM and its disk are permanently deleted. Everything is gone and cannot be recovered. **Never click "Terminate" unless you are completely finished with this VM forever.**

How to START the VM again for the next session

4

Start the instance

In EC2 Instances, select your instance, click **"Instance state"** → **"Start instance"**. Confirm. Wait 2–4 minutes for the state to change to `Running`.

5

Find the new public IP address

Each time the VM starts, it receives a **new public IP address**. Click on your instance name to open the details panel. Find **"Public IPv4 address"** and note the new value. Use this new IP in the **"Computer"** field when connecting via Remote Desktop.

> **TIP – KEEP A PERMANENT IP (ELASTIC IP)**
>
> AWS offers **Elastic IPs** — fixed IP addresses that stay the same across stop/starts. Go to EC2 → **Elastic IPs** → **Allocate Elastic IP address**, then associate it with your instance. This costs ~$0.005/hr when the instance is stopped, so release it if you stop testing for a long period.

11. Estimated Costs

Cost Item	Rate	Example: 4 hrs/day x 5 days
g4dn.xlarge (running hours)	~$0.526 / hr	**~$10.52**
EBS disk storage (60 GB, gp3)	~$0.08 / GB / month	**~$4.80 / month**
Data transfer out (RDP)	First 100 GB free	**~$0** (RDP uses minimal data)
Elastic IP (optional, while stopped)	~$0.005 / hr	**~$0.10**

Total for a typical 5-day testing sprint

4 hours of testing per day for 5 days (20 hours total) = **approximately $15–20** including compute and storage.

Accidentally leaving the VM running for 24 hours = **~$12.60 extra**. Leaving it on for a full week = **~$89**. Always stop when not in use.

TIP – SET A SPENDING ALERT

In the AWS Console: click your account name (top-right) → **Billing and Cost Management** → **Budgets** → **Create budget** → choose "Zero spend budget" (alerts you the moment any charge appears) or set a monthly budget of $30. AWS will email you when you approach the limit.

12. Troubleshooting

Cannot connect via Remote Desktop

- **Check your IP address:** Your Internet Service Provider may have assigned you a new IP. Go to `https://whatismyipaddress.com` on your local PC to find your current IP. Then in AWS go to EC2 → Security Groups → find your group → edit the inbound rule for port 3389 and update the IP address.
- **Instance not running:** Verify the instance state in EC2 is `Running` (not Stopped or Pending).
- **Wrong IP in RDP file:** If you stopped and started the VM, the IP address changes. Use the new IP from the EC2 instance details — do not reuse the old .rdp file if the IP has changed.

Application fails to start on the VM

- Open PowerShell on the VM and run: `dotnet --version`. If it shows an error, the .NET 8 Desktop Runtime is not installed. Re-run the installer from Section 6.1.
- If ServerHost crashes immediately, look for a `logs` folder inside `C:\MyLocalAssistant\ServerHost\`. Open the most recent log file to see the error message.
- If Windows shows a SmartScreen warning when you run the .exe: right-click the .exe → **Properties** → tick **"Unblock"** at the bottom → click OK.

AI inference is very slow (not using the GPU)

- Verify the NVIDIA driver: open PowerShell and run `nvidia-smi`. If it fails, reinstall the driver (Section 6.2) and restart the VM.
- Verify CUDA: in PowerShell run `nvcc --version`. If it fails, reinstall the CUDA Toolkit (Section 6.3).
- Always restart the VM after installing the driver and CUDA.
- Check the ServerHost console window — when GPU acceleration is active it will show a message containing `GPU layers` with a non-zero number.

The publish commands fail on my local PC

- Confirm the .NET 8 SDK is installed: run `dotnet --version` in PowerShell. It must show `8.x.x`. The SDK is different from the Runtime.
- If you see errors about missing NuGet packages, run `dotnet restore` first, then re-run the publish commands.
- Make sure your PowerShell prompt is in the project root folder (the one containing the `src` subfolder) before running the publish commands.

Unexpected AWS charges on my account

- In EC2 → Instances, look for any instance in the `Running` state that you did not intend to leave on. Stop it immediately.
- Go to AWS Console → **Billing** → **Bills** for a line-by-line breakdown of what was charged.
- If a large charge appears and you believe it is an error, contact AWS Support (free tier support is available for billing questions).